Suspected Adverse Drug Reaction

Case Scenario

Patient: Mr. R.K. (Case #ADR-2026-014)

Age/sex: 58-year-old/Male

Weight: 72 kg

Setting: Initially outpatient, then admitted to General Medicine.

Presenting symptoms

For three days, he had an itchy rash over much of his body, fever, and facial swelling. The rash began on his torso and spread to his arms and legs.

Medical history

Type 2 diabetes for 8 years, hypertension for 5 years, and gout diagnosed 3 weeks ago. No known medication allergies or history of liver or kidney disease. Non-smoker; occasional alcohol use.

Medication history

Medication	Dose	Indication	Started	Status
Metformin	500 mg twice daily	Diabetes	6 years ago	Continued
Amlodipine	5 mg once daily	Hypertension	4 years ago	Continued
Atorvastatin	20 mg once daily	High cholesterol	2 years ago	Continued
Allopurinol	300 mg once daily	Gout	Day 0 (4 weeks ago)	Continued until admission
Colchicine	0.5 mg twice daily	Gout flare prevention	Day 0 (4 weeks ago)	Stopped on Day 14
Paracetamol	500 mg as needed	Fever or pain	Day 24	Occasional use

Timeline

Day	Event
0	Allopurinol started at 300 mg daily without gradual dose increases; colchicine also started.
14	Colchicine stopped after the gout flare improved.
21	Mild itching began on the torso.
25	Fever of 101 F, widespread red raised rash, and facial swelling developed.
28	Presented to hospital as the rash continued to spread and worsen.

Examination

Temperature 101.4 F; heart rate 104/min; blood pressure 126/80 mmHg; oxygen saturation 97%.

There was a widespread red rash over the torso and limbs, facial swelling, tender cervical and axillary lymph nodes, and mild liver enlargement. No blisters or mucosal erosions were seen; Nikolsky sign was negative.

Investigations

Test	Result	Reference
Hemoglobin	12.8 g/dL	13-17
White blood cells	13,400/uL	4,000-11,000
Eosinophils	1,400/uL (14%)	<500/uL
Platelets	190,000/uL	150,000-400,000
ALT	210 U/L	<40
AST	165 U/L	<40
ALP	140 U/L	44-147
Bilirubin	1.4 mg/dL	<1.2

Test	Result	Reference
Creatinine	1.9 mg/dL (baseline 1.0)	0.7-1.3
Uric acid	6.2 mg/dL	3.5-7.2

Viral hepatitis tests were negative. Blood cultures showed no growth.

Treatment and outcome

Allopurinol was stopped at admission. He received intravenous fluids, skin moisturizers, antihistamines, and prednisolone 1 mg/kg/day. He improved over 10 days. Liver enzyme levels and creatinine returned to normal by week 6.

Case Analysis

1. Could this be an adverse drug reaction?

Yes. The findings strongly suggest a serious drug-related reaction. The widespread rash, fever, facial swelling, swollen lymph nodes, elevated eosinophils, and signs of liver and kidney involvement occurred after a new medicine was started. Symptoms began about three weeks after allopurinol was introduced, and the patient improved after it was stopped. Negative viral hepatitis tests and blood cultures make some other explanations less likely, but do not rule out every possible cause.

2. Most likely diagnosis

The presentation is consistent with **DRESS (drug reaction with eosinophilia and systemic symptoms)**, a severe drug reaction that can affect the skin and internal organs. A formal diagnostic score cannot be calculated from the information provided alone; additional clinical assessment may be needed to confirm the diagnosis and exclude other causes.

3. Most likely suspected medication

Allopurinol is the leading suspected drug. It was newly started, the timing fits, and it is a known cause of severe drug-related skin reactions, including DRESS. The 300 mg starting dose without gradual increases is relevant when reviewing whether the reaction might have been preventable. Prescribing guidance recommends starting at a low dose and increasing gradually, with particular care in people with impaired kidney function.

Medication	Assessment
Allopurinol	Most likely: recently started; timing fits; known association with severe skin reactions.
Colchicine	Less likely: stopped before rash and fever began; the presentation is less typical of colchicine toxicity.
Atorvastatin	Less likely: taken for 2 years without a reported recent change.
Amlodipine	Less likely: taken for 4 years without a reported recent change.
Metformin	Less likely: taken for 6 years without a reported recent change.
Paracetamol	Unlikely to have started the reaction because it was first used after itching and rash began.